ROBOTIC ARM SIMULATION — PICK-AND-PLACE

1. Aim

To design a 3D robotic arm model using OpenSCAD, define its degrees of freedom (DOF), and demonstrate a basic pick-and-place operation in which the arm picks up an object from one location, transports it, and places it at another location.

2. Software and Tools Required

- OpenSCAD for 3D modelling and animation
- OpenSCAD scripting language for defining the robot geometry and joint movements

3. Introduction

A robotic arm is a programmable mechanical system designed to perform tasks such as material handling, assembly, packaging, and object manipulation. It consists of connected links and joints that allow it to move its end-effector to a desired position.

In this project, a 3D robotic arm with a base, articulated links, wrist, and gripper was modelled using OpenSCAD. A scripted animation demonstrates the pick-and-place process by moving the gripper toward an object, lifting it, transporting it to a target location, and releasing it.

4. Robotic Arm Design

The robotic arm model consists of the following components:

1. Base: Supports the arm and provides rotational movement.
2. Shoulder joint: Connects the base to the upper arm and controls its pitch.
3. Upper arm: Main link connecting the shoulder and elbow.
4. Elbow joint: Changes the angle between the upper arm and forearm.
5. Forearm: Connects the elbow to the wrist assembly.
6. Wrist assembly: Provides additional orientation capability.
7. End-effector: The gripper used to hold and release the object.

8. Work object: The object transferred between the pickup and placement positions.

The model uses joint-angle variables and an animation parameter to demonstrate the movement sequence.

5. Degrees of Freedom (DOF)

A degree of freedom represents an independent movement or rotation available to a mechanical system.

The design represents six rotational joint axes:

Joint	Movement	Function
J1	Base rotation	Rotates the arm around its vertical axis
J2	Shoulder pitch	Raises or lowers the upper arm
J3	Elbow pitch	Bends or extends the forearm
J4	Wrist pitch	Tilts the wrist
J5	Wrist yaw	Changes the wrist's sideways orientation
J6	Wrist roll	Rotates the end-effector about its axis

The gripper also opens and closes to hold or release the object. This gripping action is an additional actuation and is not counted as one of the six rotational DOFs.

Note: The design represents six rotational axes, while the programmed pick-and-place animation may drive only the joints required for the demonstrated transfer. The remaining wrist axes can stay fixed during this basic motion.

6. Pick-and-Place Motion Simulation

The animation demonstrates the following sequence:

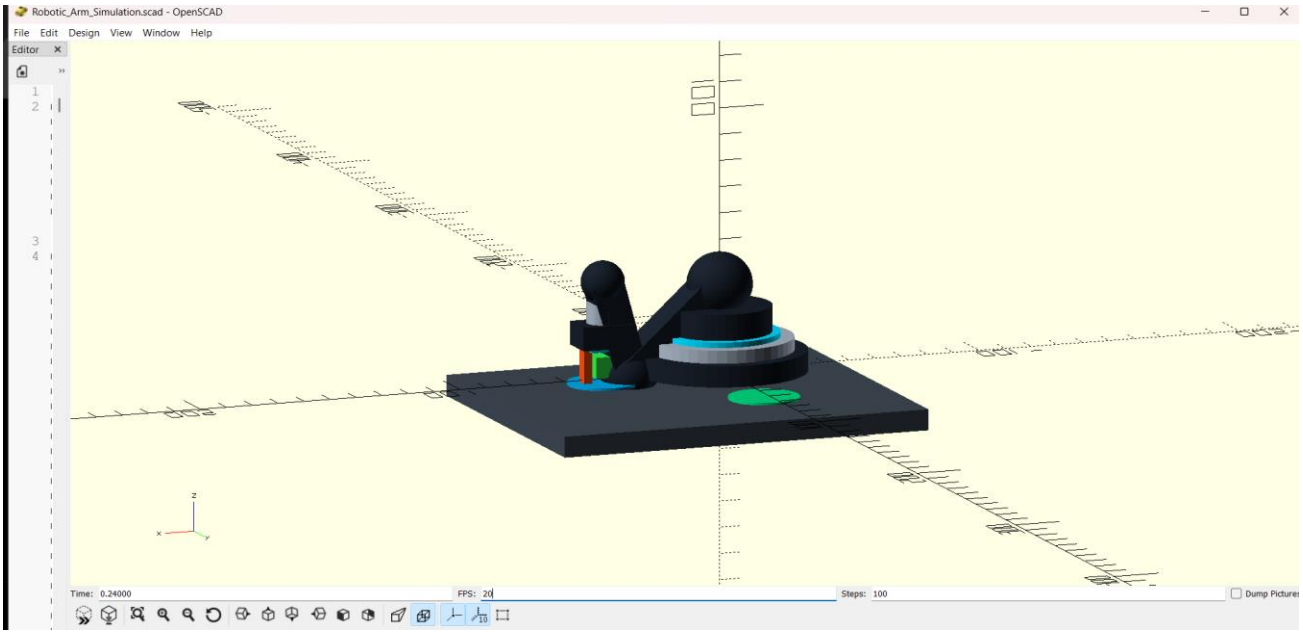
1. Initial position: The robotic arm starts in its initial configuration with the gripper positioned near the pickup area.
2. Approach: The arm moves the gripper toward the object.
3. Grasp: The gripper closes around the object.
4. Lift: The arm raises the object from the pickup location.

5. Transfer: The arm moves the object toward the destination.
6. Placement: The gripper lowers the object to the target position.
7. Release: The gripper releases the object, completing the transfer.

The animation is controlled through OpenSCAD's animation feature, which updates the model according to the animation parameter `$t`.

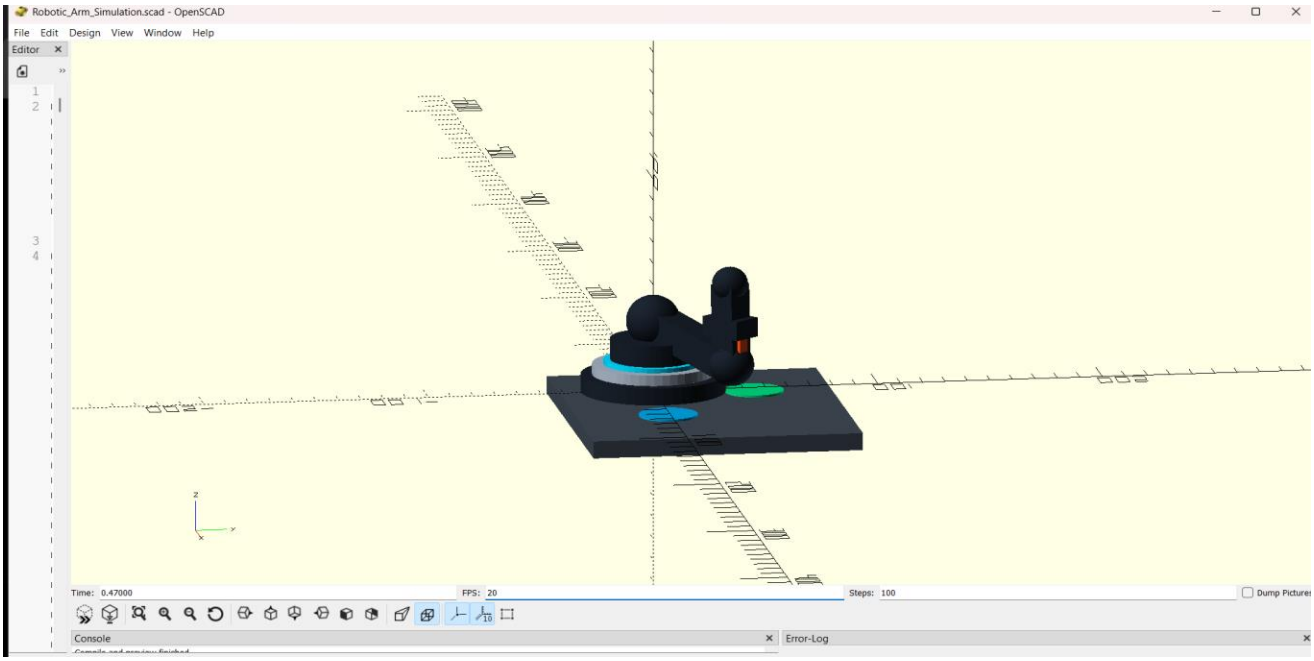
7. Simulation Screenshots

Figure 1: Robotic arm at the pickup position



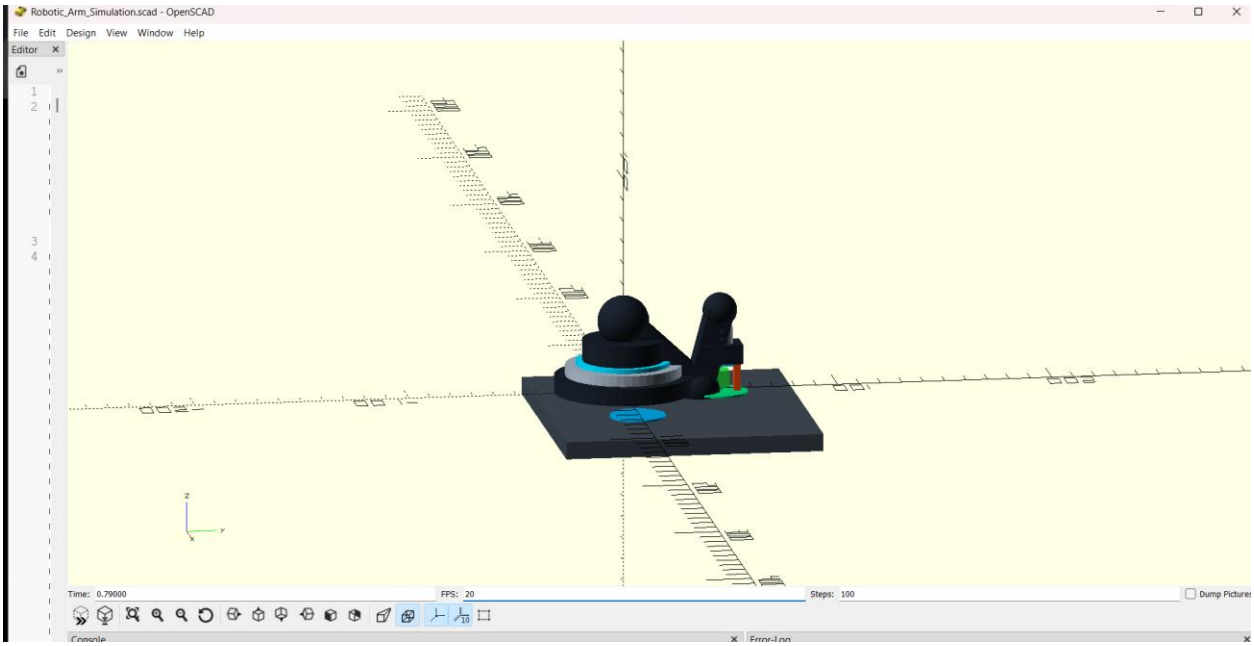
Description: The gripper approaches the object at the pickup location.

Figure 2: Robotic arm carrying the object



Description: The arm transports the object from the pickup area toward the destination.

Figure 3: Robotic arm placing the object



Description: The gripper releases the object at the destination, completing the pick-and-place sequence.

8. Results and Discussion

The 3D robotic arm was modelled successfully in OpenSCAD, and its basic pick-and-place motion was demonstrated using a scripted animation. The model illustrates the relationship between joint movement and the position of the gripper.

The simulation provides a visual understanding of robotic arm articulation, object handling, and coordinated movement. It can be extended to include more detailed inverse kinematics, collision detection, joint limits, and realistic physical simulation.

The current model is a visual and scripted demonstration. It does not, by itself, establish the physical accuracy, payload capacity, or collision-free operation of a real robotic arm.

9. Applications

Robotic arms with pick-and-place capabilities are used in:

- Industrial material handling
- Manufacturing and assembly lines
- Packaging and sorting systems
- Laboratory automation
- Warehouse automation
- Educational robotics and simulation

10. Conclusion

The project demonstrates the design of a 3D robotic arm in OpenSCAD and explains its six rotational degrees of freedom. A basic pick-and-place animation was used to illustrate how the gripper approaches an object, picks it up, moves it to a new location, and releases it.

This project provides a foundation for further study of robotic kinematics, motion planning, control systems, and industrial automation.

11. References

1. OpenSCAD Documentation — https://openscad.org/documentation.html

2. OpenSCAD User Manual — https://en.wikibooks.org/wiki/OpenSCAD_User_Manual